

Metric-like

Weak convergence

Witness function f^*

IPM Properties

- **Metric-like:**
 - Always ≥ 0 , equals 0 iff $\alpha = \beta$ (when $\mathcal{F}$ is rich enough).
 - Symmetry: $d_{\mathcal{F}}(\alpha, \beta) = d_{\mathcal{F}}(\beta, \alpha)$ (if $\mathcal{F}$ is symmetric).
 - Triangle inequality holds if $\mathcal{F}$ is convex and symmetric.
- **Weak convergence:**
 - With smooth $\mathcal{F}$, IPMs metrize weak convergence (like OT, unlike TV or KL).
 - Example: Wasserstein, MMD, Energy distance.
- **Witness function f^* :**
 - Exposes “where” α and β differ.
 - Defined only up to sign, sometimes up to additive constants.

Discrepancy Taxonomy

